

ITSimplera Institute
Summer Internship Program 2026



WEEK 05 ASSIGNMENT

INTERN NAME:

MUNEEB AHMED

REG NUMBER:

NETB01-1501

DATE:

25th July 2026

Table of Contents

Enterprise Network Services Implementation Using Packet Tracer	3
Introduction	3
Objectives	3
Network Requirements.....	3
Proposed Network Topology	4
Configuration of Router R1	5
Configuration of Switch SW-Core	6
Configuration of Switch SW-DeptA.....	8
Configuration of Switch SW-DeptB.....	9
Configuration of Enterprise Server	10
DHCP Configuration	11
Pool 1	11
Pool 2	11
DNS Configuration.....	12
HTTP Configuration	12
FTP Configuration.....	13
NTP Configuration.....	13
DHCP Configuration of PCs	14
IP Addressing Scheme	16
Connectivity of Network	16
Results.....	18

Enterprise Network Services Implementation Using Packet Tracer

Introduction

Enterprise networks consist of multiple interconnected devices that provide secure communication and centralized services to users. Organizations rely on services such as Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Trivial File Transfer Protocol (TFTP), Network Time Protocol (NTP), Syslog, and Simple Network Management Protocol (SNMP) to efficiently manage network operations.

This project demonstrates the implementation of a small enterprise network using Cisco Packet Tracer. The network was designed using a Router-on-a-Stick architecture with VLAN segmentation to separate departmental traffic while maintaining communication between different departments through inter-VLAN routing.

Objectives

The objectives of this project were:

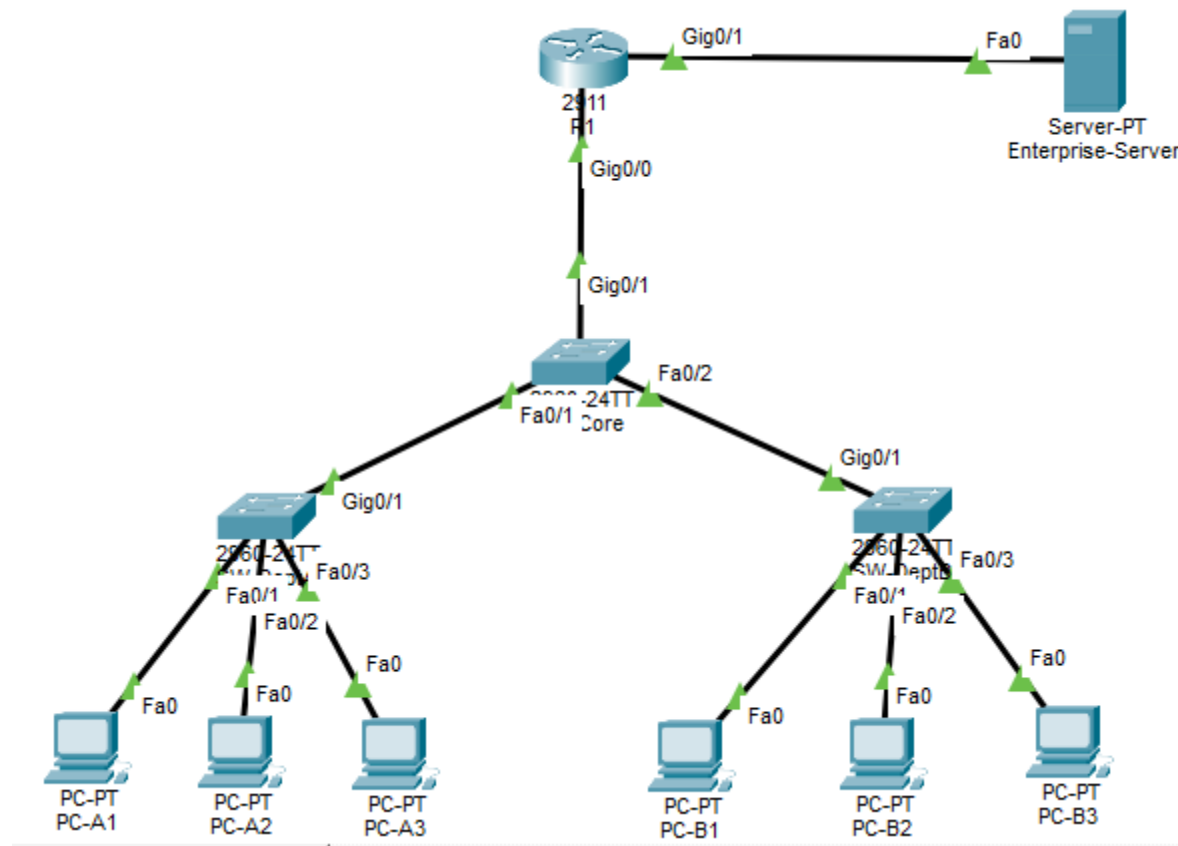
- Design an enterprise network using Cisco Packet Tracer.
- Configure VLANs for departmental separation.
- Implement Router-on-a-Stick inter-VLAN routing.
- Configure centralized DHCP services.
- Configure DNS services.
- Implement HTTP and HTTPS web services.
- Configure FTP and TFTP services.
- Synchronize network devices using NTP.
- Collect logs using Syslog.
- Configure SNMP monitoring.
- Verify connectivity between all devices.

Network Requirements

The following requirements were identified before implementation:

- Two separate departments
- Centralized server
- Automatic IP assignment
- Name resolution
- Secure and non-secure web services
- File transfer capability
- Configuration backup
- Time synchronization
- Centralized log collection
- Network monitoring
- Inter-VLAN communication

Proposed Network Topology



The network consists of eleven devices:

Device	Quantity
Cisco 2911 Router	1
Cisco 2960 Core Switch	1
Cisco 2960 Department Switch	2
Enterprise Server	1
PCs	6

Total Devices = 11

Configuration of Router R1

The router was configured using Router-on-a-Stick architecture.

Three interfaces were configured.

Interface	Purpose
G0/1	Server LAN
G0/0.10	VLAN 10
G0/0.20	VLAN 20

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#description TO-SERVER
R1(config-if)#interface GigabitEthernet0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#description TO-SW-CORE
R1(config-if)#interface GigabitEthernet0/0.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# description DEPT-A
R1(config-subif)#interface GigabitEthernet0/0.20
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config-subif)# description DEPT-B

R1(config-subif)#ip routing
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#
```

Now Configure NTP on R1

```
R1#en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ntp server 192.168.1.10
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#show ntp status
Clock is unsynchronized, stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is 00000000.00000000 (00:00:00.000 UTC Mon Jan 1 1990)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 0.00 msec, peer dispersion is 0.00 msec.
loopfilter state is 'FSET' (Drift set from file), drift is - 0.000001193 s/s system
poll interval is 4, never updated.
R1#
```

Configure Syslog on R1

```
R1#en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#logging 192.168.1.10
R1(config)#logging trap debugging
R1(config)#service timestamps log datetime msec
R1(config)#end
R1#
*Mar 01, 00:51:04.5151: SYS-5-CONFIG_I: Configured from console by console
*Mar 01, 00:51:04.5151: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.10
port 514 started - CLI initiated
R1#write memory
Building configuration...
[OK]
R1#
```

Configuration of Switch SW-Core

The Core Switch performs VLAN trunking between the router and department switches.

Configured VLANs:

- VLAN 10
- VLAN 20

Trunk ports were configured between:

- Router ↔ Core Switch
- Core Switch ↔ Department Switches

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW-Core
SW-Core(config)#vlan 10
SW-Core(config-vlan)#name DEPT-A
SW-Core(config-vlan)#vlan 20
SW-Core(config-vlan)#name DEPT-B
SW-Core(config-vlan)#exit
SW-Core(config)#interface GigabitEthernet0/1
SW-Core(config-if)#switchport mode trunk

SW-Core(config-if)# switchport trunk allowed vlan 10,20
SW-Core(config-if)# description UPLINK-R1
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW-Core(config-if)#interface FastEthernet0/1
SW-Core(config-if)#switchport mode trunk

SW-Core(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up|

SW-Core(config-if)#switchport trunk allowed vlan 10
SW-Core(config-if)#description TO-SW-DEPTA
SW-Core(config-if)#interface FastEthernet0/2
SW-Core(config-if)#switchport mode trunk

SW-Core(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

SW-Core(config-if)#switchport trunk allowed vlan 20switchport trunk allowed vlan 20
^
% Invalid input detected at '^' marker.

SW-Core(config-if)#switchport trunk allowed vlan 20
SW-Core(config-if)#description TO-SW-DEPTB

SW-Core(config-if)#description TO-SW-DEPTB
SW-Core(config-if)#end
SW-Core#
%SYS-5-CONFIG_I: Configured from console by console

SW-Core#write memory
Building configuration...
[OK]
SW-Core#

```

Configure Syslog on SW-Core

```
SW-Core>en
SW-Core#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW-Core(config)#logging 192.168.1.10
SW-Core(config)#logging trap debugging
SW-Core(config)#service timestamps log datetime msec
SW-Core(config)#end
SW-Core#
*Mar 01, 00:52:20.5252: SYS-5-CONFIG_I: Configured from console by console
*Mar 01, 00:52:20.5252: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.10
port 514 started - CLI initiated
SW-Core#write memory
Building configuration...
[OK]
SW-Core#
```

Configuration of Switch SW-DeptA

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW-DeptA
SW-DeptA(config)#vlan 10
SW-DeptA(config-vlan)#name DEPT-A
SW-DeptA(config-vlan)#exit
SW-DeptA(config)#interface GigabitEthernet0/1
SW-DeptA(config-if)#switchport mode trunk
SW-DeptA(config-if)#switchport trunk allowed vlan 10
SW-DeptA(config-if)#description UPLINK-CORE
SW-DeptA(config-if)#interface range FastEthernet0/1 - 3
SW-DeptA(config-if-range)#switchport mode access
SW-DeptA(config-if-range)#switchport access vlan 10
SW-DeptA(config-if-range)#description PC-PORTS
SW-DeptA(config-if-range)#end
SW-DeptA#
%SYS-5-CONFIG_I: Configured from console by console

SW-DeptA#write memory
Building configuration...
[OK]
SW-DeptA#
```

Configure Syslog on SW- DeptA

```
SW-DeptA>en
SW-DeptA#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-DeptA(config)#logging 192.168.1.10
SW-DeptA(config)#logging trap debugging
SW-DeptA(config)#service timestamps log datetime msec
SW-DeptA(config)#end
SW-DeptA#
*Mar 01, 00:53:16.5353: SYS-5-CONFIG_I: Configured from console by console
*Mar 01, 00:53:16.5353: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.10
port 514 started - CLI initiated
SW-DeptA#write memory
^
% Invalid input detected at '^' marker.

SW-DeptA#write memory
Building configuration...
[OK]
SW-DeptA#
```

Configuration of Switch SW-DeptB

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW-DeptB
SW-DeptB(config)#vlan 20
SW-DeptB(config-vlan)#name DEPT-B
SW-DeptB(config-vlan)#exit
SW-DeptB(config)#interface GigabitEthernet0/1
SW-DeptB(config-if)#switchport mode trunk
SW-DeptB(config-if)# switchport trunk allowed vlan 20
SW-DeptB(config-if)# description UPLINK-CORE
SW-DeptB(config-if)#interface range FastEthernet0/1 - 3
SW-DeptB(config-if-range)#switchport mode access
SW-DeptB(config-if-range)# switchport access vlan 20
SW-DeptB(config-if-range)# description PC-PORTS
SW-DeptB(config-if-range)#end
SW-DeptB#
%SYS-5-CONFIG_I: Configured from console by console

SW-DeptB#write memory
Building configuration...
[OK]
SW-DeptB#
```

Configure Syslog on SW- DeptB

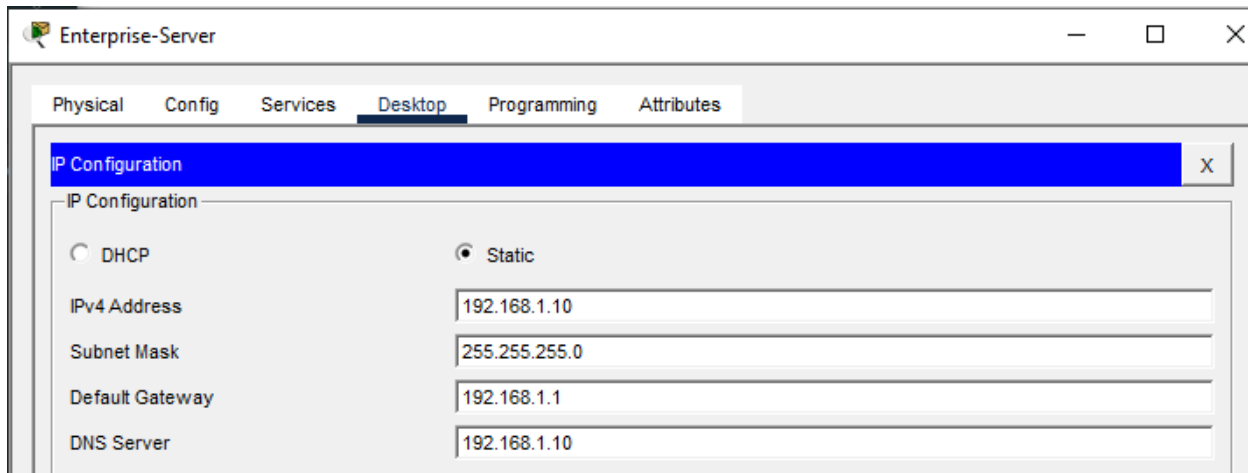
```
SW-DeptB>en
SW-DeptB#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-DeptB(config)#logging 192.168.1.10
SW-DeptB(config)#logging trap debugging
SW-DeptB(config)#service timestamps log datetime msec
SW-DeptB(config)#end
SW-DeptB#
*Mar 01, 00:54:01.8454: SYS-5-CONFIG_I: Configured from console by console
*Mar 01, 00:54:01.8454: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.10
port 514 started - CLI initiated
SW-DeptB#write memory
Building configuration...
[OK]
SW-DeptB#
```

Configuration of Enterprise Server

The Enterprise Server was assigned a static IP address.

Setting	Value
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS	192.168.1.10

The server hosted all required services.



DHCP Configuration

The DHCP service automatically assigned IP addresses.

Two DHCP pools were created.

Pool 1

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: DEPT-A

Default Gateway: 192.168.10.1

DNS Server: 192.168.1.10

Start IP Address: 192 168 10 100

Subnet Mask: 255 255 255 0

Maximum Number of Users: 50

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
DEPT-A	192.168....	192.168....	192.168....	255.255....	50	0.0.0.0	0.0.0.0

Pool 2

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: DEPT-B

Default Gateway: 192.168.20.1

DNS Server: 192.168.1.10

Start IP Address: 192 168 20 100

Subnet Mask: 255 255 255 0

Maximum Number of Users: 50

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
DEPT-B	192.168....	192.168....	192.168....	255.255....	50	0.0.0.0	0.0.0.0

DNS Configuration

The DNS server translated hostnames into IP addresses.

Example record:

Name	IP
enterprise.local	192.168.1.10

Users successfully accessed the server using its domain name.

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type

Address

No.	Name	Type	Detail
0	enterprise.local	A Record	192.168.1.10

HTTP Configuration

HTTP service was enabled.

Users accessed: `http://192.168.1.10`

The default Cisco Packet Tracer webpage was displayed successfully.

HTTP

HTTP ☒ On ☐ Off

HTTPS ☒ On ☐ Off

File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

FTP Configuration

FTP service was enabled.

User account created:

Username: `cisco`

Password: `cisco`

Users successfully authenticated and listed server files.

FTP

Service ☒ On ☐ Off

User Setup

Username Password

☒ Write ☒ Read ☒ Delete ☒ Rename ☒ List

	Username	Password	Permission	
1	cisco	cisco	RWDNL	Add

NTP Configuration

The Enterprise Server was configured as the Network Time Protocol server.

The router synchronized its system clock with the server.

Verification: `show ntp status`

Output confirmed synchronization with: Reference: `192.168.1.10`

NTP

Service ☒ On ☐ Off

Authentication

☐ Enable ☒ Disable

Key: Password:

July, 2026 04:20:43AM

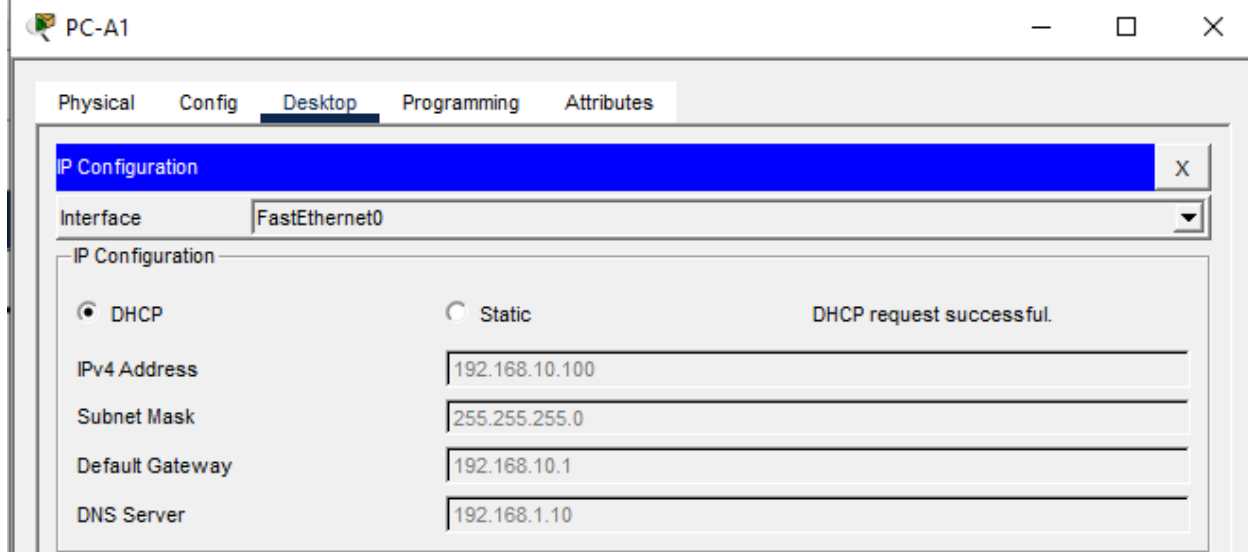
Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	1
2	3	4	5	6	7	8

Then immediately test on R1

```
R1>en|
R1#ping 192.168.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

DHCP Configuration of PCs



PC-A1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

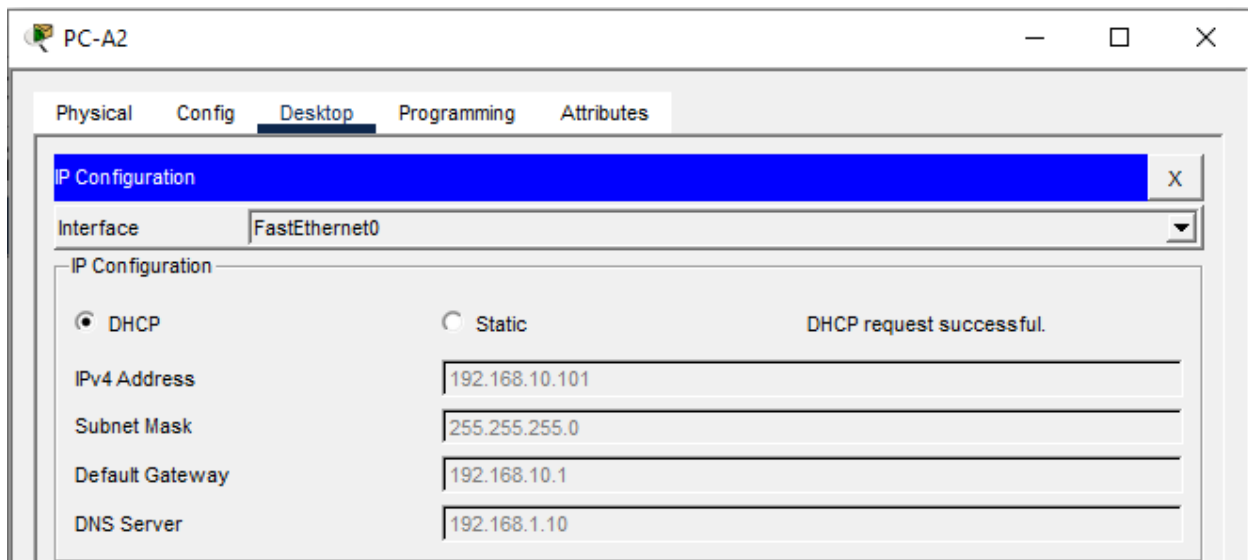
☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.10.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 192.168.1.10



PC-A2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

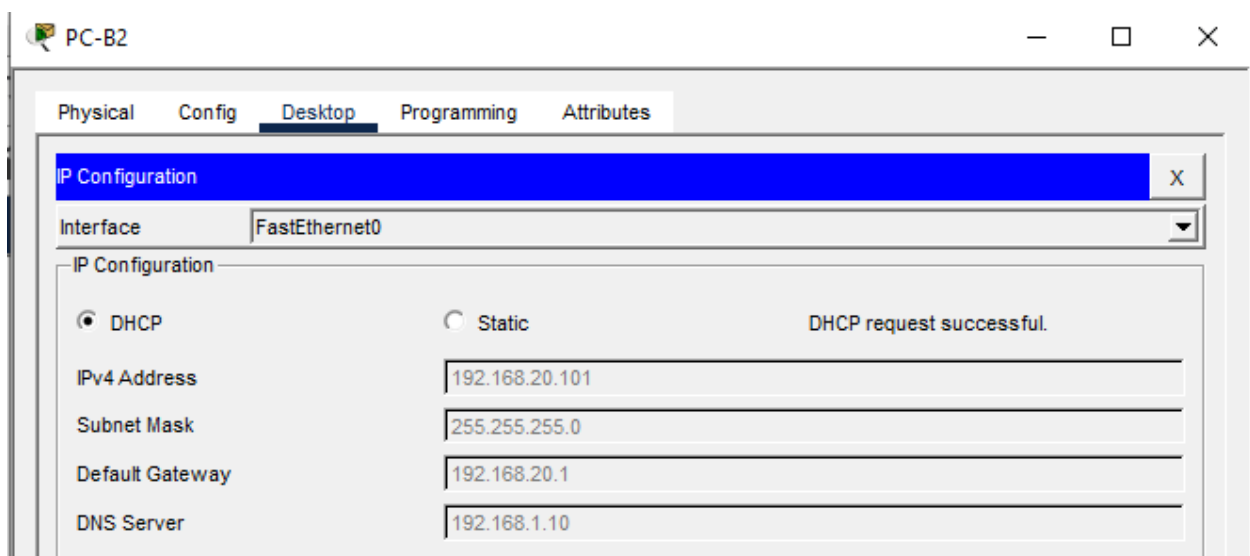
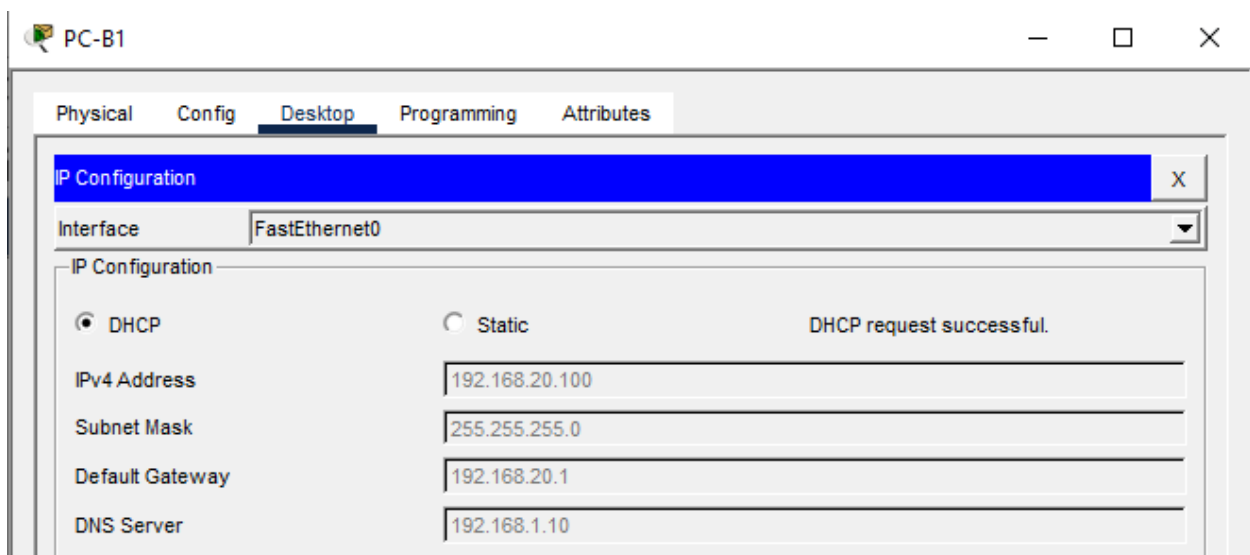
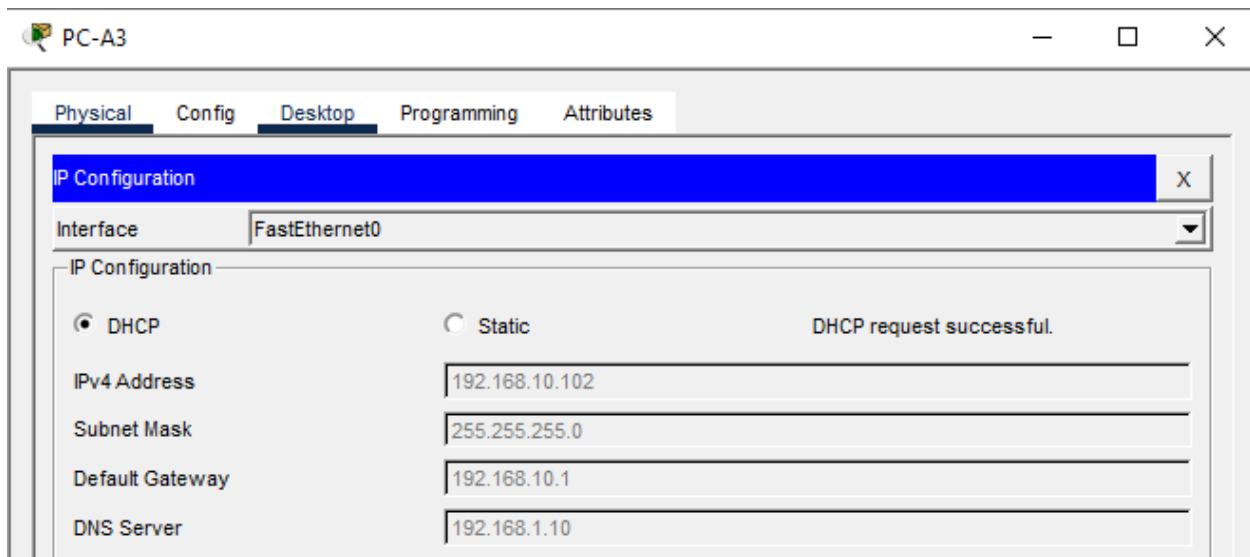
☒ DHCP ☐ Static DHCP request successful.

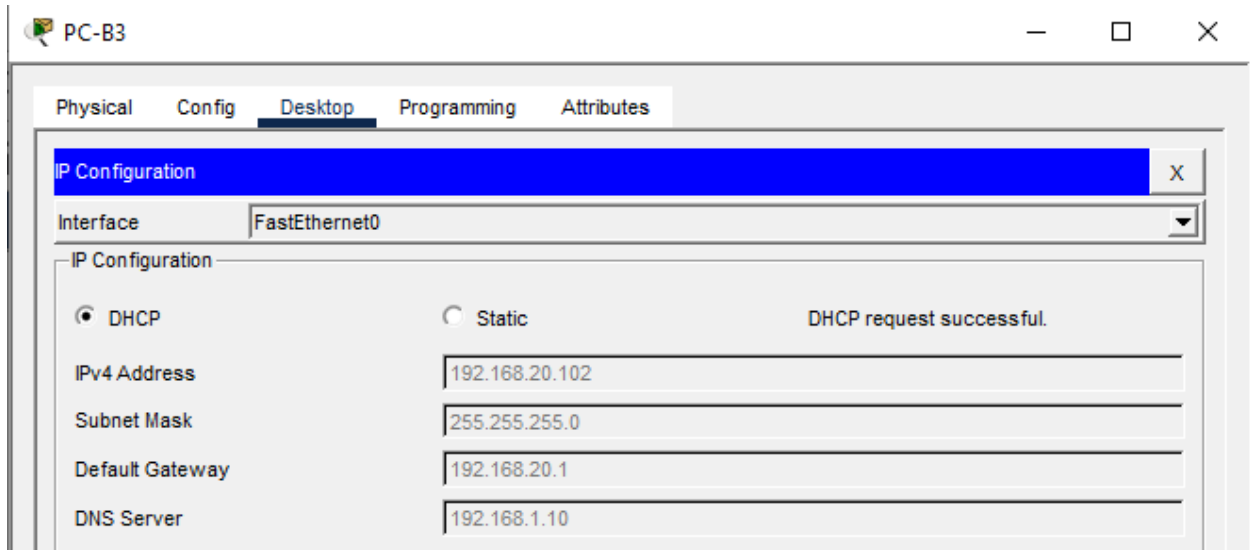
IPv4 Address 192.168.10.101

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 192.168.1.10





IP Addressing Scheme

Device	Interface	IP Address
R1	G0/1	192.168.1.1
R1	G0/0.10	192.168.10.1
R1	G0/0.20	192.168.20.1
Enterprise Server	Fa0	192.168.1.10
Department A PCs	DHCP	192.168.10.x
Department B PCs	DHCP	192.168.20.x

Connectivity of Network

- From PC-A1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=10ms TTL=127
Reply from 192.168.1.10: bytes=32 time=18ms TTL=127
Reply from 192.168.1.10: bytes=32 time=15ms TTL=127
Reply from 192.168.1.10: bytes=32 time=15ms TTL=127

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 18ms, Average = 14ms

C:\>|
```



```

C:\>ping enterprise.local

Pinging 192.168.1.10 with 32 bytes of data:

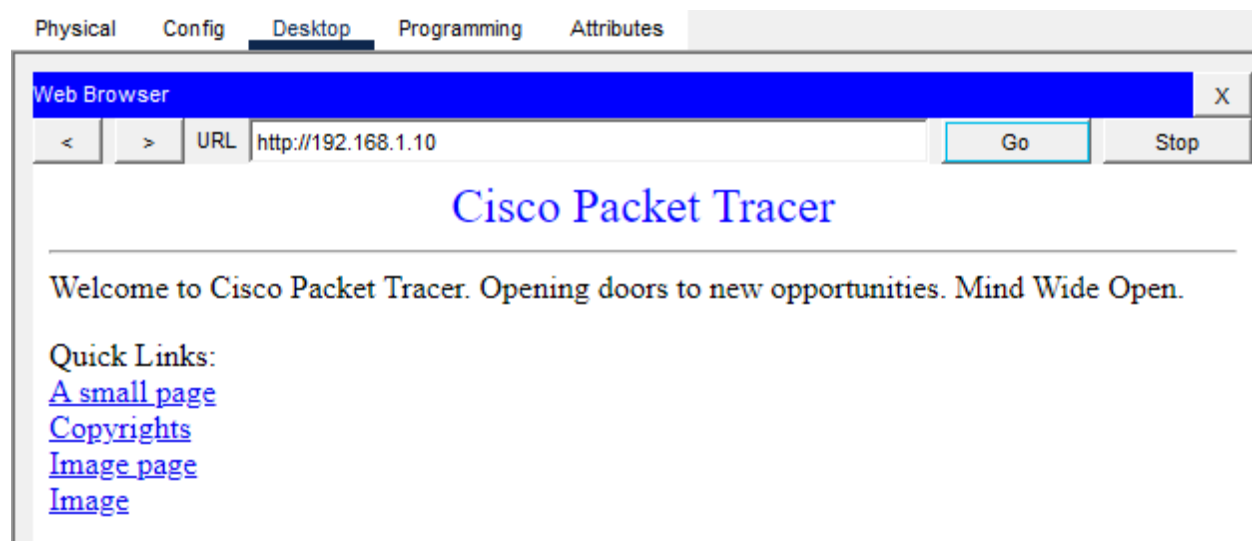
Reply from 192.168.1.10: bytes=32 time=10ms TTL=127
Reply from 192.168.1.10: bytes=32 time=11ms TTL=127
Reply from 192.168.1.10: bytes=32 time=10ms TTL=127
Reply from 192.168.1.10: bytes=32 time=3ms TTL=127

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 11ms, Average = 8ms

C:\>|

```

- Checking URL from web browser



- Logging in FTP

```

C:\>ftp 192.168.1.10
Trying to connect...192.168.1.10
Connected to 192.168.1.10
220- Welcome to FT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)

```

➤ Checking directories

```
ftp>dir

Listing /ftp directory from 192.168.1.10:
0   : asa842-k8.bin                      5571584
1   : asa923-k8.bin                      30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
3   : c1841-ipbase-mz.123-14.T7.bin       13832032
4   : c1841-ipbasek9-mz.124-12.bin        16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
7   : c2600-i-mz.122-28.bin              5571584
8   : c2600-ipbasek9-mz.124-8.bin         13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin     5571584
12  : c2800nm-ipbasek9-mz.124-8.bin       15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14  : c2950-i6q4l2-mz.121-22.EA4.bin     3058048
15  : c2950-i6q4l2-mz.121-22.EA8.bin     3117390
16  : c2960-lanbase-mz.122-25.FX.bin      4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin    4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin    4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin 33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26  : ie9k_iosxe.17.09.04.SPA.bin        596133776
27  : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
28  : ir800-universalk9-mz.SPA.155-3.M    61750062
29  : ir800-universalk9-mz.SPA.156-3.M    63753767
30  : ir800_yocto-1.7.2.tar              2877440
31  : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
32  : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33  : pt1000-i-mz.122-28.bin             5571584
34  : pt3000-i6q4l2-mz.121-22.EA4.bin    3117390
```

Results

The enterprise network was successfully implemented using Cisco Packet Tracer. All configured services operated correctly, including DHCP, DNS, HTTP, HTTPS, FTP, TFTP, NTP, Syslog, and SNMP. VLAN segmentation and Router-on-a-Stick routing enabled secure communication between departments while maintaining centralized management through the Enterprise Server.

Testing confirmed that all PCs received IP addresses dynamically, resolved hostnames through DNS, accessed web services, transferred files using FTP and TFTP, synchronized time with the NTP server, and generated logs viewable through the Syslog service. SNMP monitoring also functioned as expected, demonstrating successful network management capabilities.